PERFECT OBJECTS DESCEND ALONG LIMIT TRANSFORMATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source proves essential surjectivity by induction using the gluing lemma on a distinguished square and discharges the affine base case by the schemes result; the argument is complete

Editorial note. The label lemma-perfect-on-limit names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the descent situation of the chapter: a limit of schemes or algebraic spaces with transition maps and X the limit
- the induction principle for spaces etale over the limit
- full faithfulness of the colimit functor on perfect objects (descend-homomorphisms lemma)
- the gluing lemma for objects with maps over a distinguished square covering
- perfect objects of the derived category

Source reference. Original reference 071Q: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 402–402. Prerequisite pointer invoked at proof line(s) 6286; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09RE: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6184–6184. Prerequisite pointer invoked at proof line(s) 6249; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08HG: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3288–3288. Prerequisite pointer invoked at proof line(s) 6287; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01JB: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2206–2206. Prerequisite pointer invoked at proof line(s) 6274; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08DR: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1563–1563. Prerequisite pointer invoked at proof line(s) 6247; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09RF: sources/stacks-chow/tags.txt, source lines 12318–12318. Prerequisite pointer invoked at proof line(s) 6284; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 084R: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6175–6175. Prerequisite pointer invoked at proof line(s) 6230; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 084R the category of perfect objects of $D(\mathcal{O}_X)$ is the colimit of the categories of perfect objects of $D(\mathcal{O}_{X_i})$.*

Proof. For every quasi-compact and quasi-separated object U_0 of $(X_0)_{spaces,étale}$ consider the condition P that the functor

$$\mathrm{colim}_{i\geq 0} D_{perf}(\mathcal{O}_{U_i}) \longrightarrow D_{perf}(\mathcal{O}_U)$$

is an equivalence where $_{perf}$ indicates the full subcategory of perfect objects and where $U = X \times_{X_0} U_0$ and $U_i = X_i \times_{X_0} U_0$. We will prove P holds for every U_0 by the induction principle of Lemma 08DR. First, we observe that we already know the functor is fully faithful by Lemma 09RE. Thus it suffices to prove essential surjectivity.

We first check condition (2) of the induction principle. Thus suppose that we have an elementary distinguished square $(U_0 \subset X_0, V_0 \to X_0)$ and that P holds for U_0, V_0, and $U_0 \times_{X_0} V_0$. Let E be a perfect object of $D(\mathcal{O}_X)$. We can find $i \geq 0$ and $E_{U,i}$ perfect on U_i and $E_{V,i}$ perfect on V_i whose pullback to U and V are isomorphic to $E|_U$ and $E|_V$. Denote

$$a : E_{U,i} \to (R(X \to X_i)_* E)|_{U_i} \quad \text{and} \quad b : E_{V,i} \to (R(X \to X_i)_* E)|_{V_i}$$

the maps adjoint to the isomorphisms $L(U \to U_i)^* E_{U,i} \to E|_U$ and $L(V \to V_i)^* E_{V,i} \to E|_V$. By fully faithfulness, after increasing i, we can find an isomorphism $c : E_{U,i}|_{U_i \times_{X_i} V_i} \to E_{V,i}|_{U_i \times_{X_i} V_i}$ which pulls back to the identifications

$$L(U \to U_i)^* E_{U,i}|_{U \times_X V} \to E|_{U \times_X V} \to L(V \to V_i)^* E_{V,i}|_{U \times_X V}.$$

Apply Lemma 01JB to get an object E_i on X_i and a map $d : E_i \to R(X \to X_i)_* E$ which restricts to the maps a and b over U_i and V_i. Then it is clear that E_i is perfect and that d is adjoint to an isomorphism $L(X \to X_i)^* E_i \to E$.

Finally, we check condition (1) of the induction principle, in other words, we check the lemma holds when X_0 is affine. This follows from the case of schemes, see Derived Categories of Schemes, Lemma 09RF. To see this use the equivalence of Lemma 071Q and use the translation of Lemma 08HG. □